

Market Concentration and the Flattening of the Phillips Curve

A Theory-Motivated, Semi-Structural State-Space Assessment of the HSA Mechanism
Estimation Results Summary

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Hybrid NKPC specification: the inflation equation adds an intercept c and lagged inflation $\alpha_b \pi_{t-1}$ to the forward-looking term $\alpha E_t \pi_{t+1}$.

1 Data Specifications

The empirical specifications vary along two dimensions: the inflation–expectation pair and the measure of the Phillips-curve forcing variable. Table 1 summarizes the alternatives considered throughout the estimation.

Table 1: Empirical Data Specifications

Panel A. Inflation–Expectation Pairs	
Inflation Measure	Expected Inflation Measure
PPI	GDP price index inflation forecast
Headline CPI	CPI inflation forecast
Core CPI	CPI inflation forecast
Panel B. Phillips-Curve Forcing Variables	
Measure	Interpretation
Inverse markup	Proxy for real marginal cost
Negative unemployment gap	Labor-market slack measure
Beveridge–Nelson output gap	BN-decomposition output gap
Hodrick–Prescott output gap	HP-filter output gap

The results below report the primary specification (PPI, inverse markup) for each case. The remaining inflation and forcing combinations enter the full robustness grid.

2 Case 1: Quarterly Joint Estimation with Capital IQ Competition Data

2.1 Coefficient Table

Table 2: Case 1: posterior mean and 95% credible interval across Models 0–4 (ppi, inverse markup).

Model	c	α	α_b	κ_0	δ	θ_0	γ
CES	-0.335 [-1.930, 1.100]	0.343 [-0.278, 0.982]	0.799 [0.688, 0.910]	-0.398 [-1.113, 0.286]	—	—	—
Slope	-0.372 [-1.919, 1.150]	0.341 [-0.289, 0.987]	0.801 [0.690, 0.911]	-0.393 [-1.130, 0.340]	0.036 [-0.288, 0.357]	—	—
Direct	-0.352 [-1.910, 1.184]	0.348 [-0.327, 0.968]	0.800 [0.686, 0.904]	-0.379 [-1.141, 0.379]	—	0.007 [-0.292, 0.285]	—
Dynamic	-0.339 [-1.913, 1.144]	0.347 [-0.283, 1.004]	0.796 [0.689, 0.903]	-0.387 [-1.121, 0.374]	—	0.005 [-0.293, 0.295]	-0.011 [-0.191, 0.190]
Joint	-0.423 [-1.961, 1.142]	0.372 [-0.266, 1.041]	0.803 [0.698, 0.909]	-0.372 [-1.107, 0.342]	0.036 [-0.291, 0.374]	-0.002 [-0.290, 0.302]	-0.018 [-0.218, 0.194]

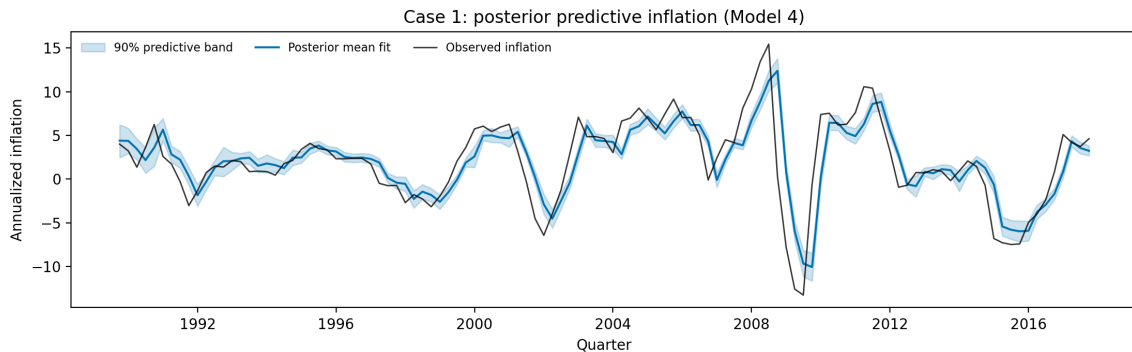
2.1.1 Marginal Likelihoods

Models are compared by the log marginal likelihood $\log m(y)$ (higher is better; Laplace–Metropolis estimator with the exact Kalman integrated likelihood for the linear Models 0–2 and a particle-filter likelihood for the bilinear Models 3–4) and by the Widely Applicable Information Criterion (WAIC; lower is better). The directional probabilities in the same table summarize the RQ1–RQ4 posterior evidence.

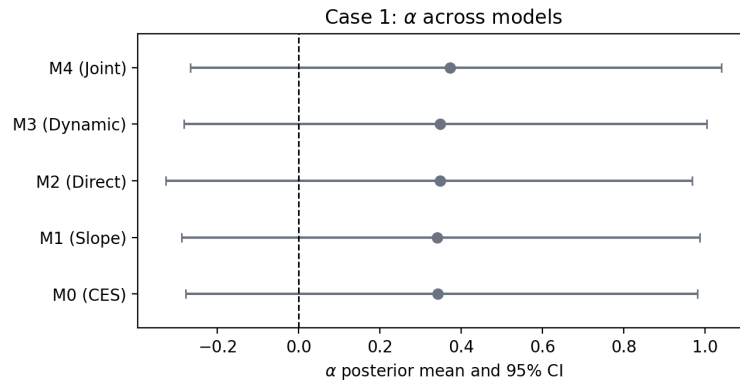
Table 3: Case 1: model comparison by WAIC (lower is better) and log marginal likelihood $\log m(y)$ (higher is better), with the posterior directional probabilities addressing RQ1–RQ4: $P(\delta > 0)$, $P(\theta_0 > 0)$, $P(\gamma > 0)$, and the joint HSA probability $P(\delta > 0, \theta_0 > 0, \gamma > 0)$. $\hat{R}$ is the largest structural-coefficient convergence statistic; † marks $\hat{R} > 1.1$ (not fully converged).

Model	$\hat{R}$	WAIC	$\log m(y)$	$P(\delta > 0)$	$P(\theta_0 > 0)$	$P(\gamma > 0)$	$P(\text{joint})$
CES	1.00	565.4	-477.0	—	—	—	—
Slope	1.00	565.9	-477.2	0.598	—	—	—
Direct	1.00	565.7	-476.5	—	0.532	—	—
Dynamic	1.01	565.7	-765.7	—	0.520	0.447	—
Joint	1.00	566.3	-621.2	0.599	0.482	0.431	0.110

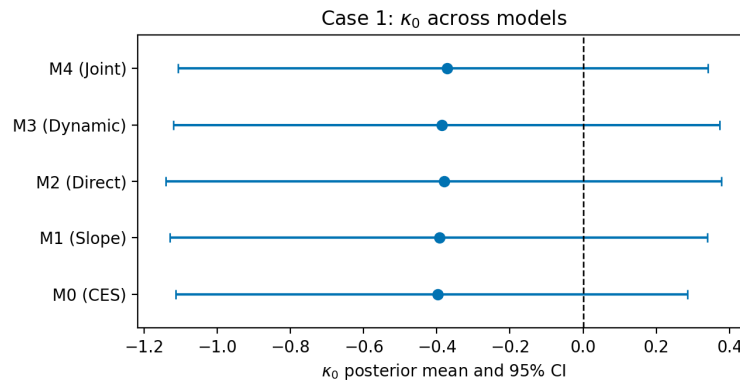
2.1.2 Predictive Distributions



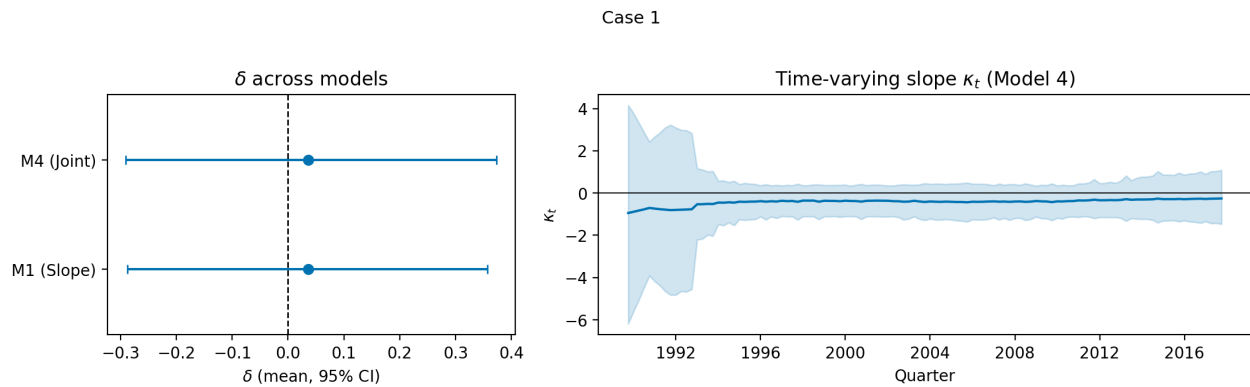
2.2 Coefficient: α



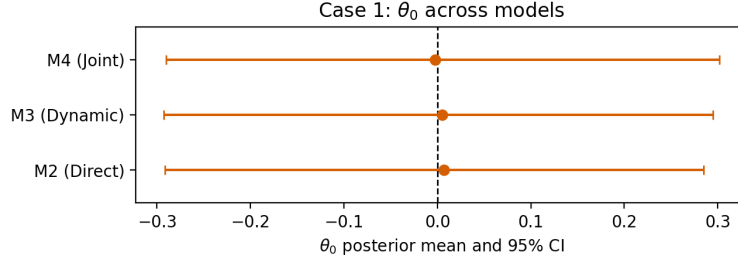
2.3 Coefficient: κ



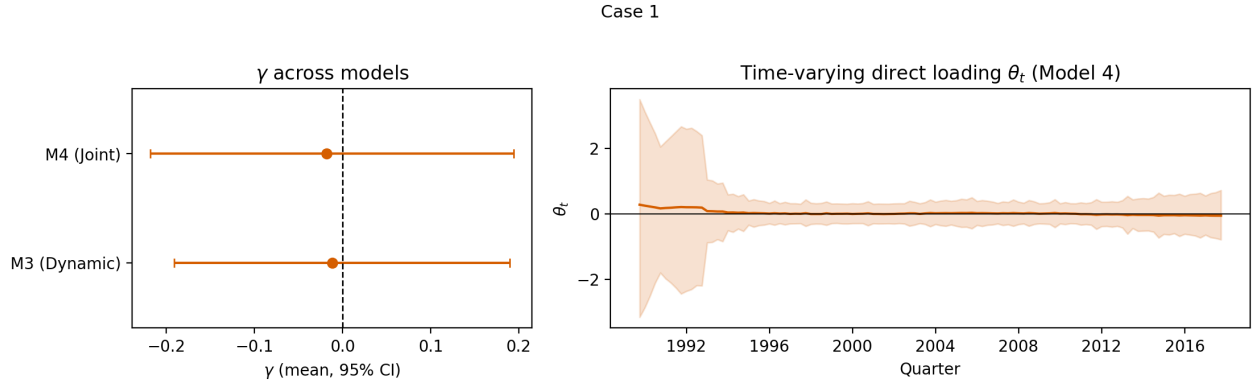
2.3.1 Coefficient: δ and Time-Varying κ_t



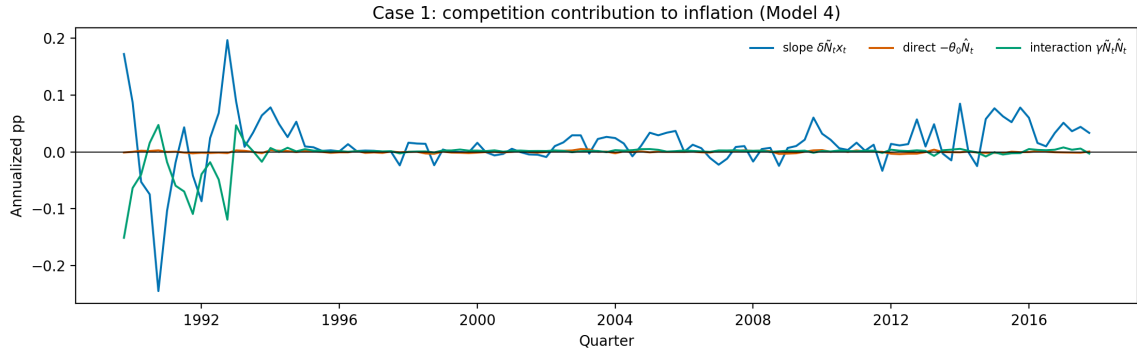
2.4 Coefficient: θ



2.4.1 Coefficient: γ and Time-Varying θ_t



2.4.2 Decomposed Competition



2.5 Robustness Across Specifications

The detailed results above use the primary specification (PPI, inverse markup). Table 4 summarizes the joint-model directional evidence across every estimated inflation and forcing combination.

Table 4: Case 1: robustness of the joint HSA directional evidence (Model 4) across all estimated inflation/forcing specifications. $\hat{R}$ is the largest structural-coefficient convergence statistic; † marks $\hat{R} > 1.1$ (directional probabilities for those rows are not fully converged).

Specification	$\hat{R}$	$\log m(y)$	$P(\delta > 0)$	$P(\theta_0 > 0)$	$P(\gamma > 0)$	$P(\text{joint})$
PPI / inv. markup (firm weighted)	1.00	-621.2	0.599	0.482	0.431	0.110
PPI / inv. markup (revenue weighted)	1.01	-497.6	0.570	0.582	0.450	0.145
PPI / inv. markup (sic4 firm)	1.01	-236.6	0.572	0.516	0.557	0.175
PPI / inv. markup (sic4 rev)	1.01	-291.2	0.517	0.687	0.572	0.204

3 Case 2: Quarterly Joint Estimation with Interpolated Gustavo Competition Data

3.1 Coefficient Table

Table 5: Case 2: posterior mean and 95% credible interval across Models 0–4 (ppi, inverse markup).

Model	c	α	α_b	κ_0	δ	θ_0	γ
CES	-0.112 [-0.862, 0.662]	0.207 [-0.011, 0.429]	0.800 [0.716, 0.883]	-0.010 [-0.579, 0.558]	—	—	—
Slope	-0.120 [-0.942, 0.654]	0.209 [-0.012, 0.424]	0.800 [0.710, 0.886]	0.018 [-0.693, 0.710]	0.035 [-0.287, 0.359]	—	—
Direct	-0.081 [-0.917, 0.800]	0.214 [-0.018, 0.445]	0.797 [0.701, 0.887]	0.020 [-0.541, 0.593]	—	0.049 [-0.134, 0.225]	—
Dynamic	-0.126 [-0.911, 0.693]	0.211 [-0.009, 0.426]	0.800 [0.715, 0.889]	0.008 [-0.584, 0.601]	—	0.017 [-0.199, 0.208]	0.001 [-0.107, 0.117]
Joint	-0.122 [-0.955, 0.692]	0.207 [-0.022, 0.423]	0.804 [0.717, 0.889]	0.018 [-0.531, 0.581]	0.100 [-0.182, 0.381]	0.009 [-0.195, 0.218]	0.001 [-0.111, 0.116]

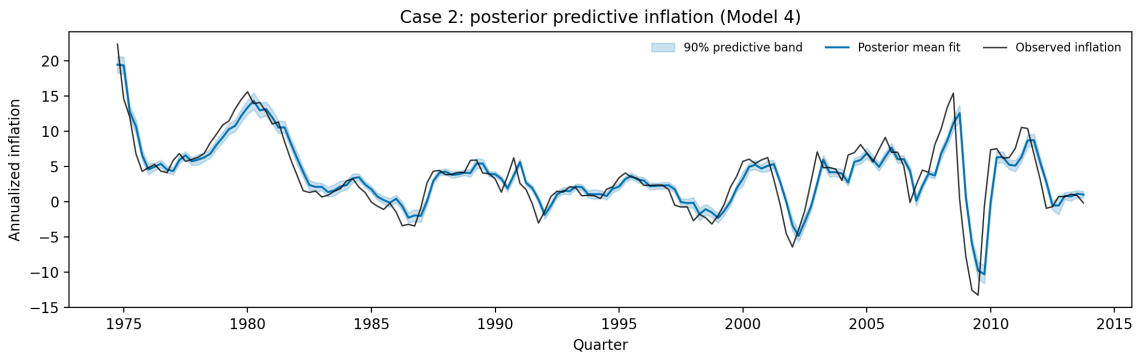
3.1.1 Marginal Likelihoods

Models are compared by the log marginal likelihood $\log m(y)$ (higher is better; Laplace–Metropolis estimator with the exact Kalman integrated likelihood for the linear Models 0–2 and a particle-filter likelihood for the bilinear Models 3–4) and by the Widely Applicable Information Criterion (WAIC; lower is better). The directional probabilities in the same table summarize the RQ1–RQ4 posterior evidence.

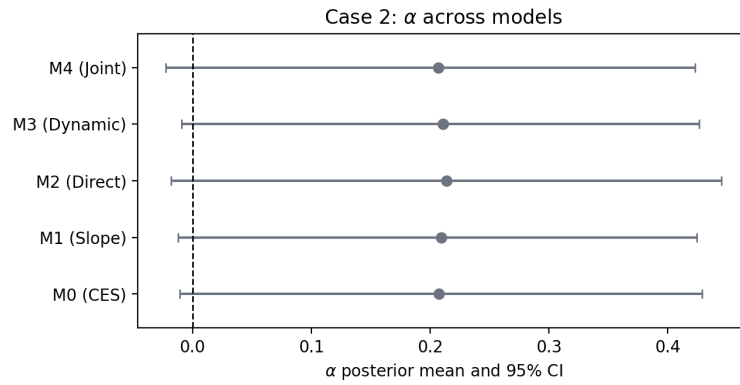
Table 6: Case 2: model comparison by WAIC (lower is better) and log marginal likelihood $\log m(y)$ (higher is better), with the posterior directional probabilities addressing RQ1–RQ4: $P(\delta > 0)$, $P(\theta_0 > 0)$, $P(\gamma > 0)$, and the joint HSA probability $P(\delta > 0, \theta_0 > 0, \gamma > 0)$. $\hat{R}$ is the largest structural-coefficient convergence statistic; † marks $\hat{R} > 1.1$ (not fully converged).

Model	$\hat{R}$	WAIC	$\log m(y)$	$P(\delta > 0)$	$P(\theta_0 > 0)$	$P(\gamma > 0)$	$P(\text{joint})$
CES	1.00	747.9	-305.8	—	—	—	—
Slope	1.00	747.5	-305.9	0.583	—	—	—
Direct	1.00	748.1	-306.2	—	0.718	—	—
Dynamic	1.01	747.5	-306.5	—	0.584	0.503	—
Joint	1.01	747.1	-308.4	0.750	0.545	0.502	0.211

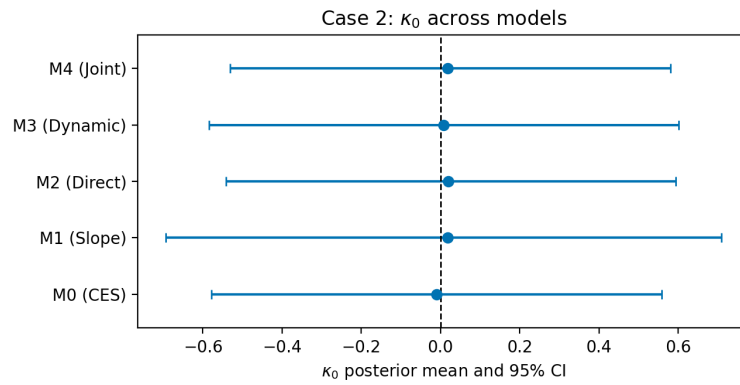
3.1.2 Predictive Distributions



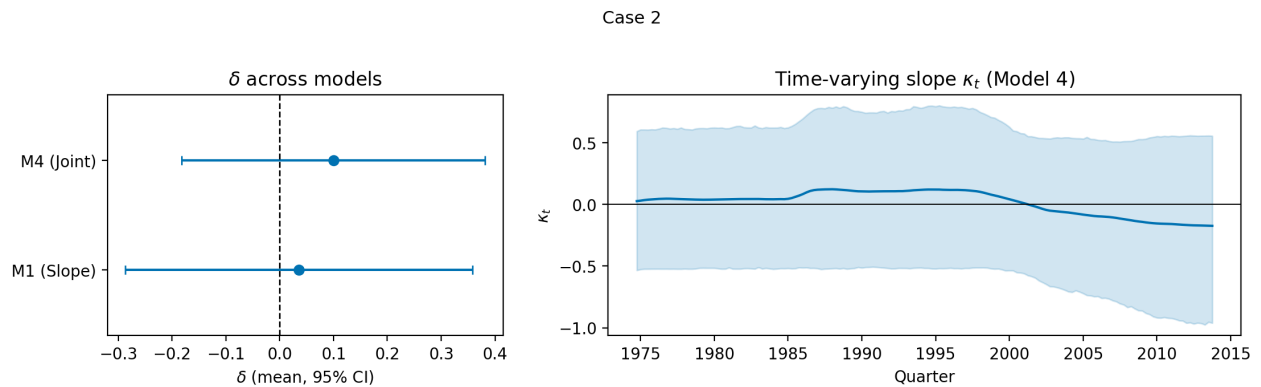
3.2 Coefficient: α



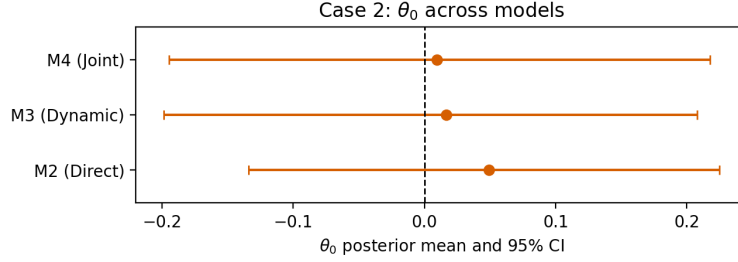
3.3 Coefficient: κ



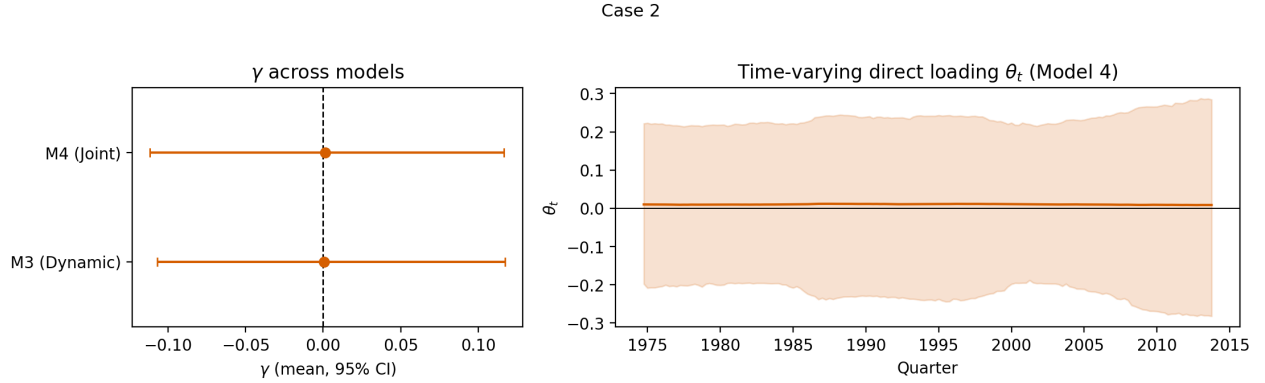
3.3.1 Coefficient: δ and Time-Varying κ_t



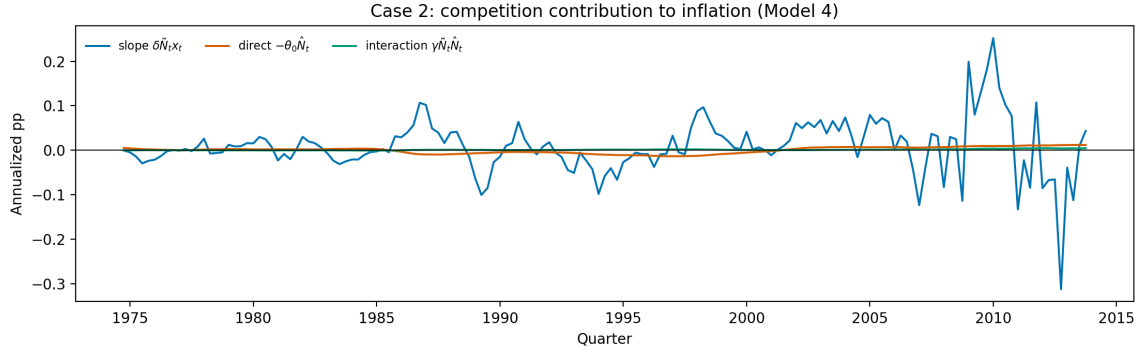
3.4 Coefficient: θ



3.4.1 Coefficient: γ and Time-Varying θ_t



3.4.2 Decomposed Competition



3.5 Robustness Across Specifications

The detailed results above use the primary specification (PPI, inverse markup). Table 7 summarizes the joint-model directional evidence across every estimated inflation and forcing combination.

Table 7: Case 2: robustness of the joint HSA directional evidence (Model 4) across all estimated inflation/forcing specifications. $\hat{R}$ is the largest structural-coefficient convergence statistic; † marks $\hat{R} > 1.1$ (directional probabilities for those rows are not fully converged).

Specification	$\hat{R}$	$\log m(y)$	$P(\delta > 0)$	$P(\theta_0 > 0)$	$P(\gamma > 0)$	$P(\text{joint})$
PPI / inv. markup	1.01	-308.4	0.750	0.545	0.502	0.211

4 Case 3: Mixed Frequency Joint Estimation with Gustavo Competition Data

4.1 Coefficient Table

Table 8: Case 3: posterior mean and 95% credible interval across Models 0–4 (ppi, inverse markup).

Model	c	α	α_b	κ_0	δ	θ_0	γ
CES	-0.118 [-0.927, 0.676]	0.206 [-0.005, 0.429]	0.801 [0.711, 0.889]	-0.003 [-0.566, 0.614]	—	—	—
Slope	-0.109 [-0.909, 0.676]	0.205 [-0.019, 0.432]	0.801 [0.706, 0.895]	0.013 [-0.668, 0.733]	0.029 [-0.276, 0.326]	—	—
Direct	-0.150 [-1.083, 0.686]	0.228 [0.007, 0.463]	0.796 [0.709, 0.883]	0.011 [-0.583, 0.589]	—	0.043 [-0.141, 0.226]	—
Dynamic	-0.119 [-0.899, 0.677]	0.213 [-0.005, 0.441]	0.800 [0.710, 0.886]	-0.008 [-0.586, 0.589]	—	0.015 [-0.183, 0.209]	0.003 [-0.111, 0.115]
Joint	-0.125 [-0.896, 0.677]	0.214 [0.002, 0.438]	0.798 [0.695, 0.886]	0.014 [-0.543, 0.591]	0.059 [-0.266, 0.352]	0.036 [-0.155, 0.217]	0.002 [-0.114, 0.113]

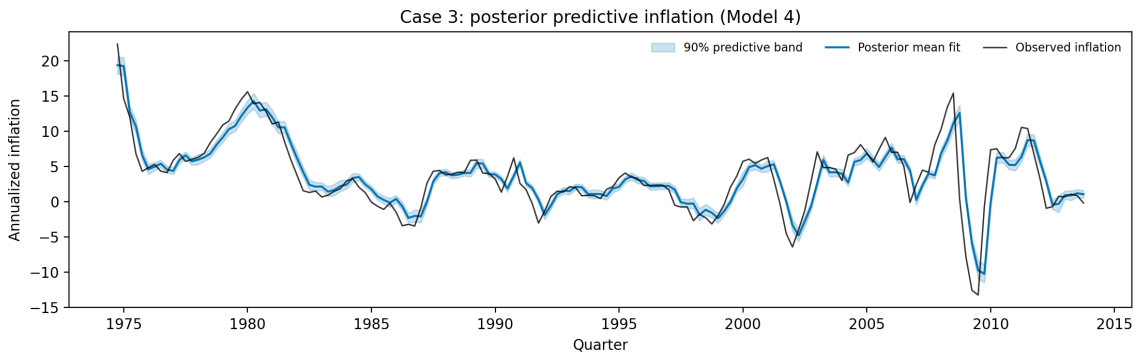
4.1.1 Marginal Likelihoods

Models are compared by the log marginal likelihood $\log m(y)$ (higher is better; Laplace–Metropolis estimator with the exact Kalman integrated likelihood for the linear Models 0–2 and a particle-filter likelihood for the bilinear Models 3–4) and by the Widely Applicable Information Criterion (WAIC; lower is better). The directional probabilities in the same table summarize the RQ1–RQ4 posterior evidence.

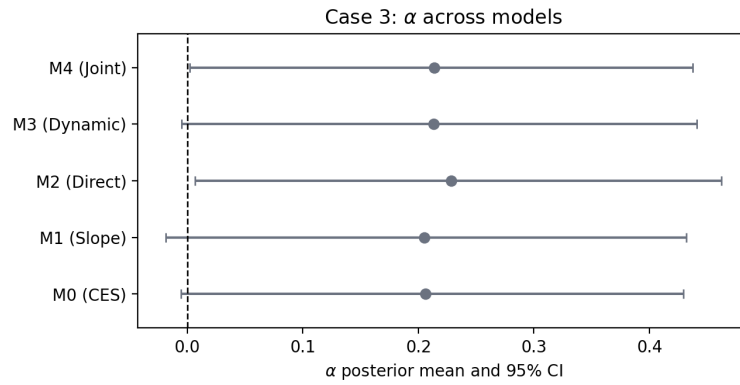
Table 9: Case 3: model comparison by WAIC (lower is better) and log marginal likelihood $\log m(y)$ (higher is better), with the posterior directional probabilities addressing RQ1–RQ4: $P(\delta > 0)$, $P(\theta_0 > 0)$, $P(\gamma > 0)$, and the joint HSA probability $P(\delta > 0, \theta_0 > 0, \gamma > 0)$. $\hat{R}$ is the largest structural-coefficient convergence statistic; † marks $\hat{R} > 1.1$ (not fully converged).

Model	$\hat{R}$	WAIC	$\log m(y)$	$P(\delta > 0)$	$P(\theta_0 > 0)$	$P(\gamma > 0)$	$P(\text{joint})$
CES	1.00	748.5	-407.3	—	—	—	—
Slope	1.00	747.9	-408.4	0.574	—	—	—
Direct	1.00	747.5	-408.2	—	0.678	—	—
Dynamic	1.02	747.5	-409.4	—	0.555	0.500	—
Joint	1.01	747.5	-409.2	0.660	0.655	0.506	0.229

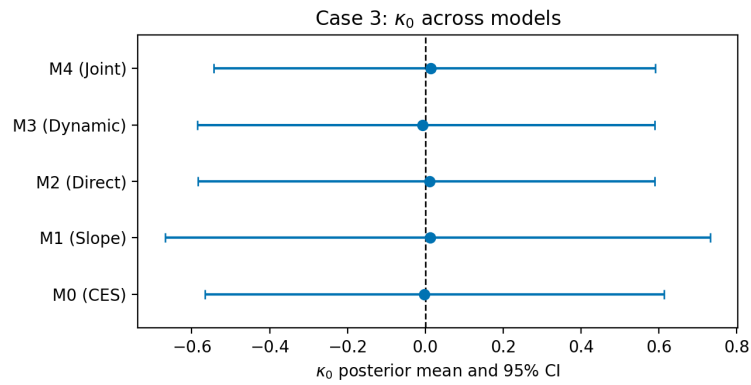
4.1.2 Predictive Distributions



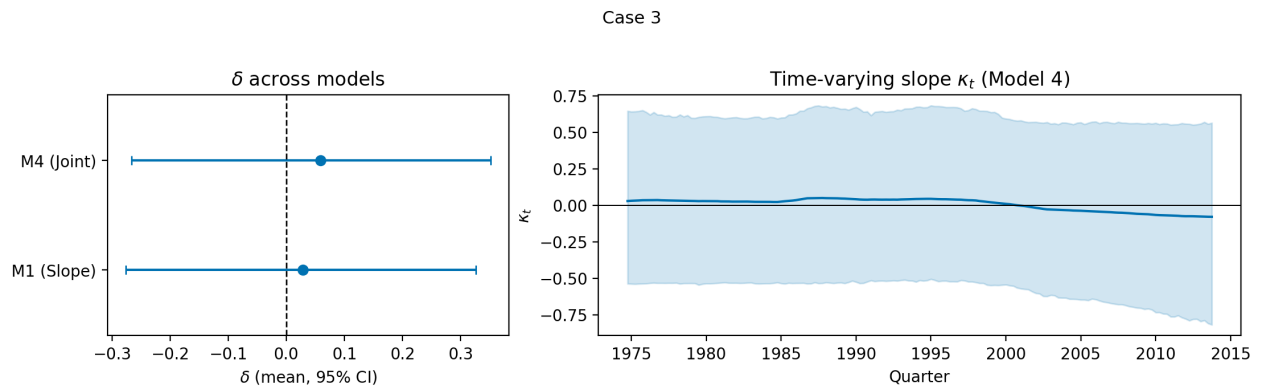
4.2 Coefficient: α



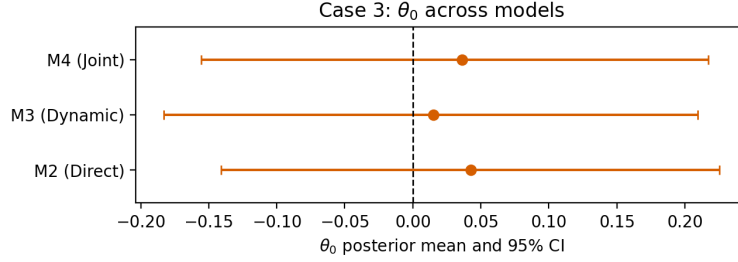
4.3 Coefficient: κ



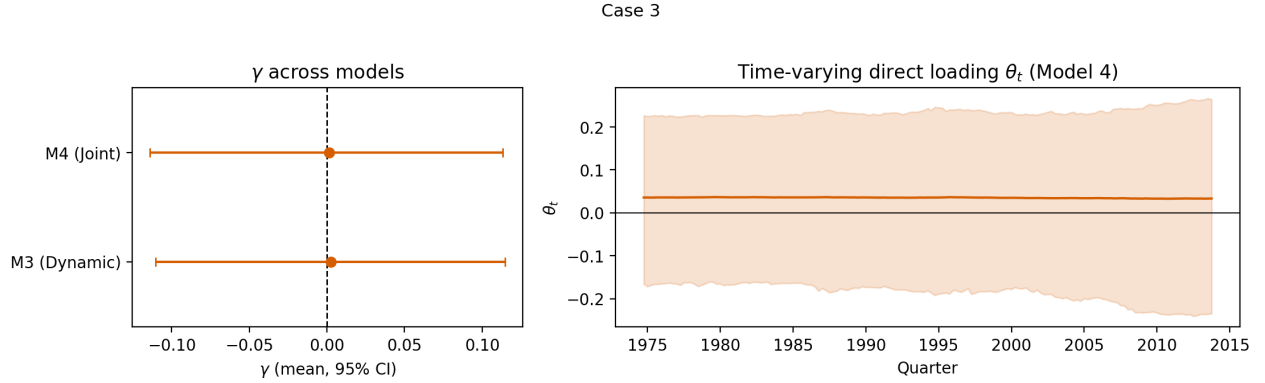
4.3.1 Coefficient: δ and Time-Varying κ_t



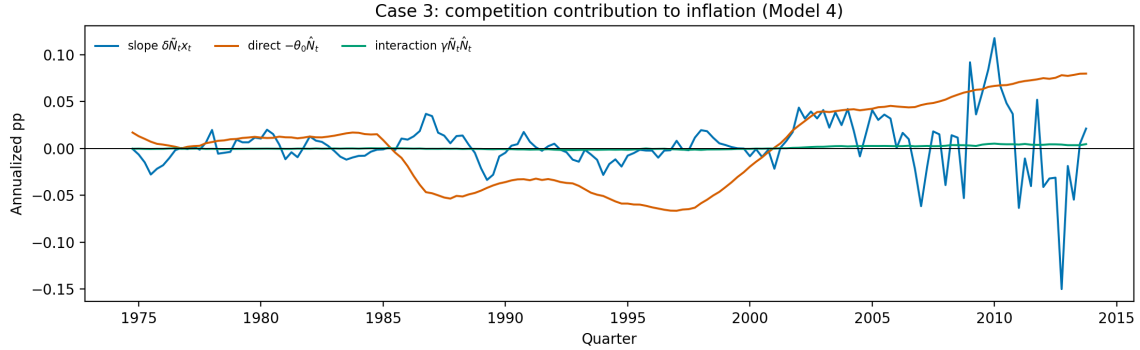
4.4 Coefficient: θ



4.4.1 Coefficient: γ and Time-Varying θ_t



4.4.2 Decomposed Competition



4.5 Robustness Across Specifications

The detailed results above use the primary specification (PPI, inverse markup). Table 10 summarizes the joint-model directional evidence across every estimated inflation and forcing combination.

Table 10: Case 3: robustness of the joint HSA directional evidence (Model 4) across all estimated inflation/forcing specifications. $\hat{R}$ is the largest structural-coefficient convergence statistic; † marks $\hat{R} > 1.1$ (directional probabilities for those rows are not fully converged).

Specification	$\hat{R}$	$\log m(y)$	$P(\delta > 0)$	$P(\theta_0 > 0)$	$P(\gamma > 0)$	$P(\text{joint})$
PPI / inv. markup	1.01	-409.2	0.660	0.655	0.506	0.229

5 Case 4: Mixed Frequency Joint Estimation with Gustavo Competition Data and Number of Establishments

5.1 Coefficient Table

Table 11: Case 4: posterior mean and 95% credible interval across Models 0–4 (ppi, inverse markup).

Model	c	α	α_b	κ_0	δ	θ_0	γ
CES	-0.488 [-2.377, 1.496]	0.511 [-0.335, 1.344]	0.778 [0.646, 0.919]	-0.250 [-1.102, 0.618]	—	—	—
Slope	-0.523 [-2.504, 1.427]	0.533 [-0.290, 1.426]	0.776 [0.628, 0.914]	-0.251 [-1.117, 0.588]	0.030 [-0.262, 0.329]	—	—
Direct	-0.495 [-2.346, 1.250]	0.516 [-0.302, 1.358]	0.776 [0.631, 0.913]	-0.238 [-1.087, 0.600]	—	0.017 [-0.193, 0.212]	—
Dynamic	-0.480 [-2.280, 1.310]	0.514 [-0.310, 1.290]	0.777 [0.637, 0.908]	-0.243 [-1.050, 0.607]	—	-0.002 [-0.206, 0.203]	0.000 [-0.113, 0.109]
Joint	-0.559 [-2.454, 1.326]	0.534 [-0.299, 1.394]	0.781 [0.646, 0.914]	-0.236 [-1.032, 0.654]	0.067 [-0.238, 0.354]	-0.000 [-0.197, 0.211]	0.001 [-0.105, 0.112]

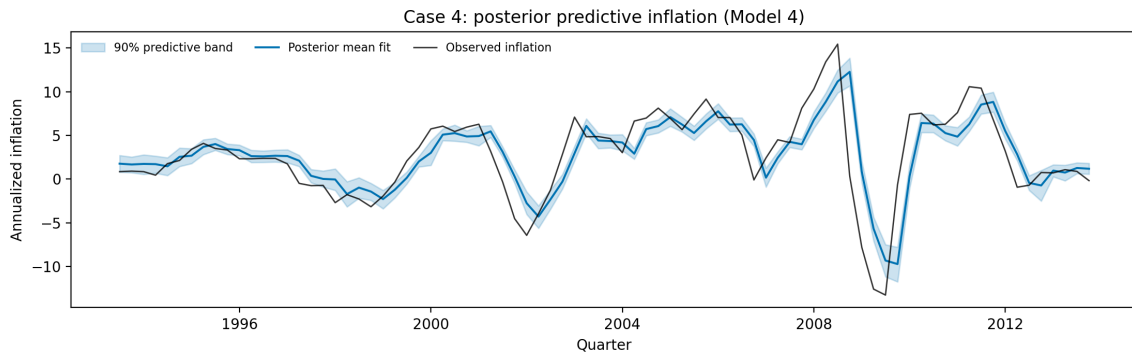
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Models are compared by the log marginal likelihood $\log m(y)$ (higher is better; Laplace–Metropolis estimator with the exact Kalman integrated likelihood for the linear Models 0–2 and a particle-filter likelihood for the bilinear Models 3–4) and by the Widely Applicable Information Criterion (WAIC; lower is better). The directional probabilities in the same table summarize the RQ1–RQ4 posterior evidence.

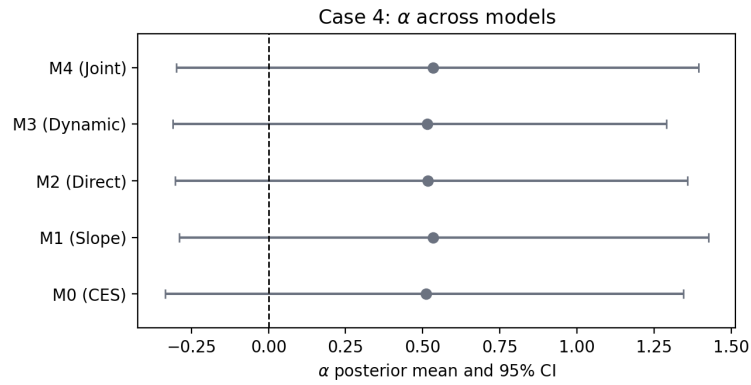
Table 12: Case 4: model comparison by WAIC (lower is better) and log marginal likelihood $\log m(y)$ (higher is better), with the posterior directional probabilities addressing RQ1–RQ4: $P(\delta > 0)$, $P(\theta_0 > 0)$, $P(\gamma > 0)$, and the joint HSA probability $P(\delta > 0, \theta_0 > 0, \gamma > 0)$. $\hat{R}$ is the largest structural-coefficient convergence statistic; † marks $\hat{R} > 1.1$ (not fully converged).

Model	$\hat{R}$	WAIC	$\log m(y)$	$P(\delta > 0)$	$P(\theta_0 > 0)$	$P(\gamma > 0)$	$P(\text{joint})$
CES	1.00	426.0	-234.3	—	—	—	—
Slope	1.01	426.4	-235.6	0.584	—	—	—
Direct	1.01	426.4	-234.8	—	0.585	—	—
Dynamic	1.01	426.4	-238.6	—	0.497	0.501	—
Joint	1.00	425.9	-240.5	0.698	0.493	0.526	0.184

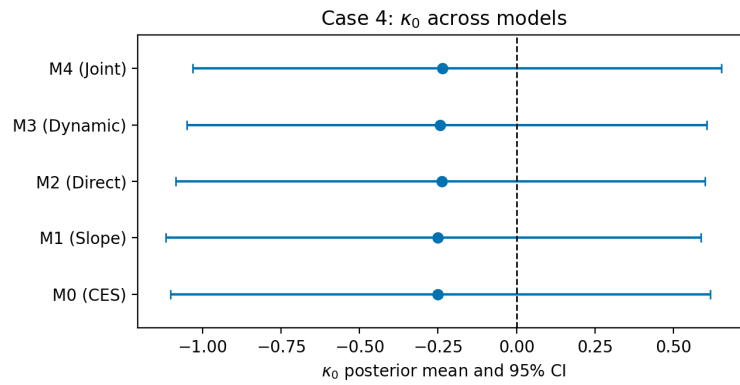
5.1.2 Predictive Distributions



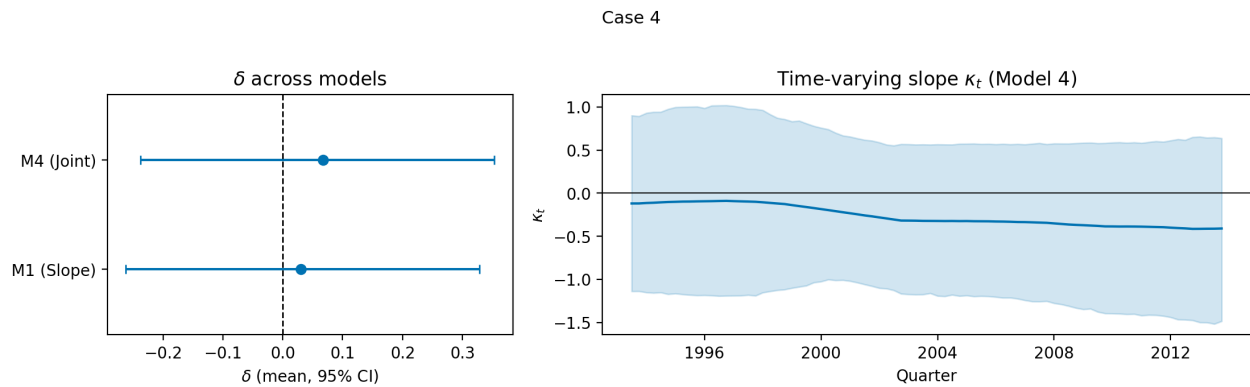
5.2 Coefficient: α



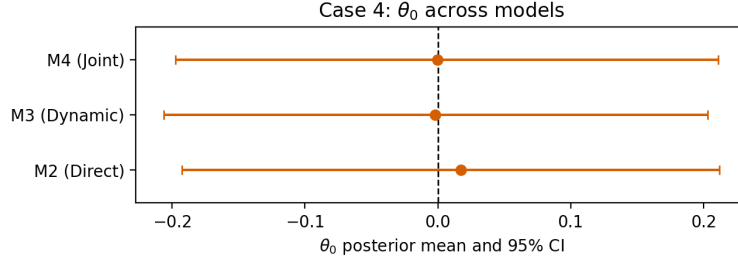
5.3 Coefficient: κ



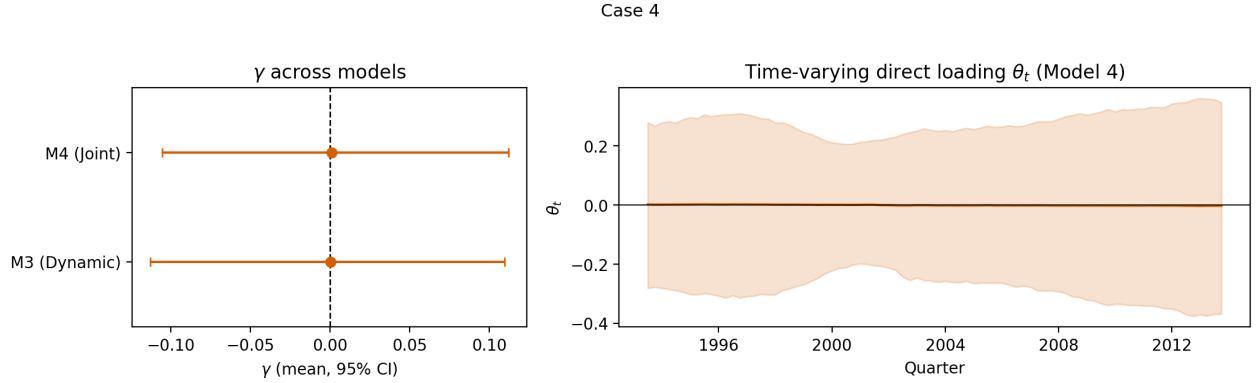
5.3.1 Coefficient: δ and Time-Varying κ_t



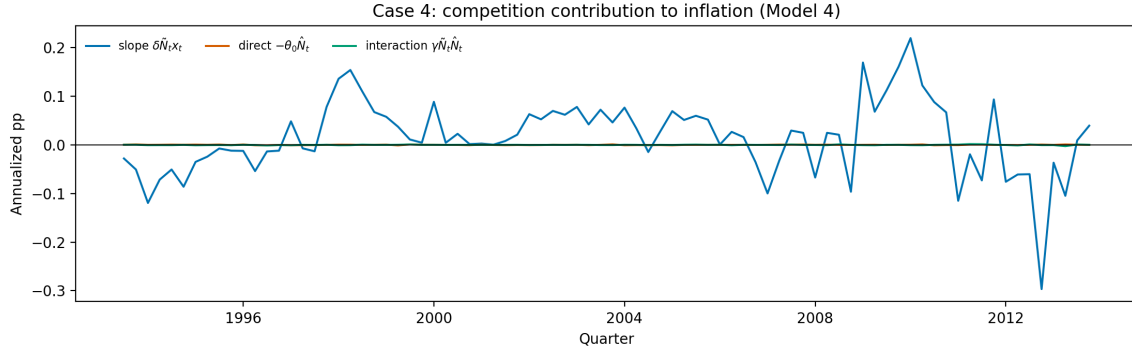
5.4 Coefficient: θ



5.4.1 Coefficient: γ and Time-Varying θ_t



5.4.2 Decomposed Competition



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The detailed results above use the primary specification (PPI, inverse markup). Table 13 summarizes the joint-model directional evidence across every estimated inflation and forcing combination.

Table 13: Case 4: robustness of the joint HSA directional evidence (Model 4) across all estimated inflation/forcing specifications. $\hat{R}$ is the largest structural-coefficient convergence statistic; † marks $\hat{R} > 1.1$ (directional probabilities for those rows are not fully converged).

Specification	$\hat{R}$	$\log m(y)$	$P(\delta > 0)$	$P(\theta_0 > 0)$	$P(\gamma > 0)$	$P(\text{joint})$
PPI / inv. markup	1.00	-240.5	0.698	0.493	0.526	0.184